Deep Learning for Downscaling Geostationary Satellite Measurements for Wildfire Detection

Luke Taulbee, Yifan Yang, Haonan Chen, *Senior Member, IEEE*, Zhengwei Yang, *Member, IEEE*, Qi Chen, Zhe Li, and Rick Mueller

***Abstract*—Despite the availability of various satellite-based fire products, near-real-time detection of fire perimeters remains challenging due to the inherent limitations in the spatio-temporal resolution of current satellite observations. For example, the Advanced Baseline Imager (ABI) onboard the Geostationary Operational Environmental Satellite (GOES) offers frequent observations, yet its coarse spatial resolution limits the accuracy of fire delineation. In contrast, low-Earth orbit (LEO) satellites such as Suomi-NPP, NOAA-20, and NOAA-21, equipped with the Visible Infrared Imaging Radiometer Suite (VIIRS), provide high spatial resolution but significantly low temporal coverage. To overcome these limitations, this article presents a generative adversarial network (GAN)-based framework that downscales GOES ABI imagery to a spatial resolution of 375 m in support of wildfire detection. To train the proposed GAN, VIIRS observations are used as ground-truth labels. In the inference, the downscaled GOES ABI images are processed through an active fire detection algorithm to produce the final wildfire product. Experimental results demonstrate that the proposed framework can successfully enhance the spatial resolution of wildfire products using downscaled GOES observations to 375 m while preserving its high temporal frequency of 10 min, allowing more precise and timely wildfire detection. The proposed framework introduces a novel combination of GAN-based super-resolution and operational wildfire detection, offering a practical approach to enhance geostationary satellite observations for timely fire monitoring.**

***Index Terms*—Deep learning (DL), generative adversarial network (GAN), Geostationary Operational Environmental Satellite (GOES), satellite remote sensing, Visible Infrared Imaging Radiometer Suite (VIIRS), wildfire detection.**

I. Introduction

WILDFIRE risk has increased significantly due to overgrown forests, prolonged droughts, and rising global temperatures. In 2024 alone, wildfires burned approximately 8.9 million acres across the United States [1]. These events cause long-term ecological damage by leaving behind ash and partially burned material, which hinders vegetation regrowth and increases the risk of mudslides during rain [2]. More critically, wildfires pose severe threats to land, homes, infrastructure, and human life. For example, the 2023 Maui fires caused 115 fatalities and 5.52 billion dollars in damages after burning 3200 acres [3]. These impacts underscore the urgent need for improved wildfire detection systems to enhance emergency response and mitigation efforts.

Existing wildfire detection systems, including ground-based sensors and aerial platforms, offer distinct advantages but also come with inherent limitations. Aerial systems, such as drones, helicopters, and fixed-wing aircraft, enable high-resolution and localized monitoring, often in near-real time, making them particularly valuable in remote and inaccessible regions [4], [5], [6]. Drones, in particular, offer autonomous operation capabilities and are equipped with infrared sensors for effective night-time detection. Despite these advantages, aerial platforms face several operational constraints, including limited spatial coverage, sensitivity to adverse weather conditions, short flight endurance, and the need for substantial personnel and maintenance support [4], [6].

Satellite-based systems, on the other hand, provide broad and continuous coverage for wildfire monitoring from space. This research focuses on two primary orbital types: low-Earth orbit (LEO) and geostationary Earth orbit (GEO). LEO satellites, such as Landsat, Suomi-NPP, and NOAA-20, are equipped with high-resolution sensors capable of detecting small fires and hotspots with fine spatial detail. However, their utility for near-real-time wildfire monitoring is constrained by limited repeat cycles [7], [8]. For instance, LEO satellites equipped with Visible Infrared Imaging Radiometer Suite (VIIRS) with a repeat cycle of 16 days, meaning one of the VIIRS satellites will fully cycle and take an image of the same geographic location approximately every 16 days, making them less effective for tracking rapidly evolving fire events.

In contrast, GEO satellites, such as the Geostationary Operational Environmental Satellite (GOES) series, carry sensors such as the Advanced Baseline Imager (ABI), which provide high-frequency observations up to every 10 min over fixed geographic regions. While this high temporal resolution enables continuous monitoring, the relatively coarse spatial resolution (~2 km) limits their effectiveness in detecting small or low-intensity fires. Both GEO and LEO satellite sensors

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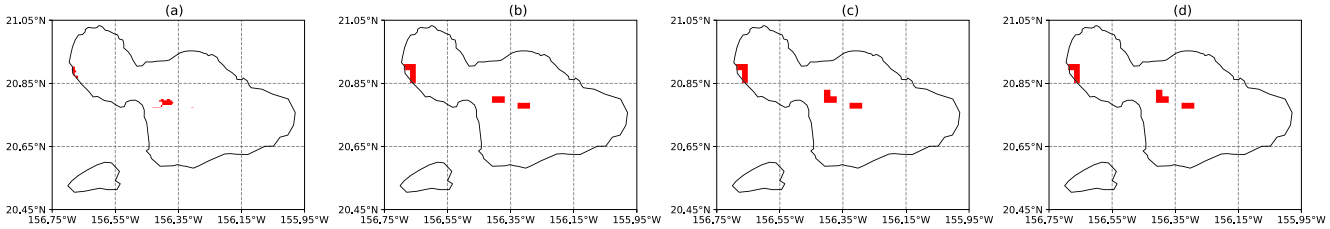


Fig. 1. Various fire detection products on August 9, 2023, near 12:00 UTC over Maui Island, Hawaii: (a) VIIRS active fire product and (b)–(d) GOES FDC product.

collect radiance data across multiple channels, which are subsequently converted into brightness temperatures (BTs) for thermal analysis. Fire detection algorithms, such as VIIRS's active fire product and fire detection and characterization (FDC) system based on GOES ABI observations, leverage this thermal information to identify active fires and thermal anomalies [9], [10], [11]. For example, Fig. 1 presents the satellite-based fire detection results from (a) VIIRS and (b)–(d) GOES on August 9, 2023. While VIIRS provides high spatial resolution, the GOES products offer much higher temporal frequency, with updates available every 10 min. These space-based systems substantially reduce the dependence on ground-based methods and significantly enhance early fire detection at regional to global scales.

To overcome the inherent spatio-temporal tradeoffs of both LEO and GEO satellite systems, this study introduces FireGAN, a generative adversarial network (GAN) model for high-resolution wildfire detection. Our novel FireGAN model enhances GOES ABI imagery by downscaling its native 2-km spatial resolution to the finer 375 m, while preserving its high temporal frequency (10-min intervals). By effectively combining the temporal advantage of GOES with the downscaled spatial details, FireGAN allows high-resolution imagery that supports near-real-time wildfire detection and monitoring.

This study conducts a comprehensive performance assessment of the proposed FireGAN and wildfire detection results over the selected study domain. The key contributions of this work are summarized as follows.

1) We propose FireGAN, designed to downscale GOES observations from 2 km to 375 m, producing imagery with both high spatial and temporal resolution.
2) By leveraging the enhanced temporal frequency of GOES and the spatial fidelity achieved through downscaling, we generate a wildfire detection product with 375-m resolution at 10-min intervals.
3) Comprehensive evaluations are conducted to assess the accuracy and reliability of FireGAN's wildfire detection outputs, demonstrating its effectiveness for near-real-time wildfire monitoring.

Overall, FireGAN provides an entirely novel approach for satellite-based wildfire detection by integrating GAN-based downscaling with an active fire detection product, bridging the gap between spatial resolution and temporal coverage in geostationary imagery.

The remainder of the article is organized as follows. Section II provides a brief review of related work on satellite observation downscaling. Section III describes the datasets and the study domain used in this research. Section IV details the proposed methodology, including an overview of the framework, data preprocessing steps, the GAN-based downscaling, and the performance evaluation metrics. Section V presents the experimental setups and results, followed by the discussion in Section VI. Finally, Section VII concludes the article with a summary of key findings and future research directions.

II. Related Work

Deep learning (DL) approaches for image super-resolution have been extensively studied, with significant applications in remote sensing and wildfire detection. A prominent contribution is the enhanced super-resolution GAN (ESRGAN) developed by Wang et al. [12]. ESRGAN builds upon the original GAN architecture by introducing residual-in-residual dense blocks (RRDBs), which enhance hierarchical feature extraction and improve reconstruction fidelity. It also adopts the relativistic average GAN (RaGAN) framework [13], enabling the discriminator to estimate the relative realism between real and generated images, thereby enhancing perceptual quality and reducing visual artifacts. In addition, ESRGAN utilizes perceptual loss computed from preactivated VGG-19 features, aiding in preserving fine textures and structural details in the output images. While ESRGAN produces visually compelling results, its focus on perceptual quality rather than pixel-level fidelity may limit its suitability for remote sensing applications, where precise numerical accuracy is critical for tasks such as quantitative wildfire monitoring and detection. To address super-resolution from a probabilistic perspective, Cai et al. [14] proposed a multistage variational autoencoder (VAE) that leverages latent variables to preserve spatial information and employs residual blocks for improved reconstruction quality. Despite these innovations, the model's relatively shallow architecture and suboptimal loss function design often result in blurred outputs, limiting its effectiveness for applications requiring high-fidelity spatial details.

In the context of remote sensing, numerous works have utilized DL for remote sensing downscaling [15], [16], [17], [18]. For example, Shao et al. [15] developed a coupled sparse autoencoder (CSAE) augmented with a sparse coefficients of residuals (SCORs) module to enhance high-frequency details in satellite imagery. The model leverages sparse representations and residual-based refinement to learn a mapping between low- to high-resolution features. While CSAE demonstrates strong performance in preserving perceptual quality, its reliance on relatively high-resolution datasets (1–18 m)

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and emphasis on perceptual quality over pixel-level accuracy constrain its applicability to wildfire monitoring tasks, which demand precise numerical fidelity for effective detection and analysis. For wildfire-specific applications, Badhan et al. [18] designed an autoencoder (AE) to downscale GOES-17 imagery from 2-km resolution to 375 m. While the encoder–decoder architecture captures general image features, the absence of adversarial or perceptual loss components hinders its ability to reconstruct fine details, such as sharp fire boundaries. As a result, the model achieves only moderate performance in terms of intersection over union (IoU), with scores ranging from 0.12 to 0.14.

Next, considering other approaches such as diffusion models, such as denoising diffusion probabilistic models (DDPMs) and related conditional diffusion frameworks, have shown strong performance in image generation, restoration, and super-resolution by modeling the data distribution through a gradual denoising process. Notable works include SR3, which applies conditional diffusion for image super-resolution and variants tailored for remote sensing restoration and enhancement. While these diffusion-based methods offer state-of-the-art results in perpetual quality and uncertainty quantification, this study requires direct mapping from GOES ABI and VIIRS imagery, rather than modeling a stochastic denoising process. This allows our approach to produce high-resolution imagery efficiently at operational geostationary cadence, which is essential for near-real-time wildfire detection. We refer readers to diffusion-based super-resolution literature [19], [20], [21], [22].

Finally, recent advances in remote sensing change detection and segmentation have emphasized the use of strong feature representations and effective adaptation under limited labeled data. Several studies have explored integrating large pretrained vision models, such as the segment anything model (SAM), with task-specific feature interaction mechanisms to improve robustness and generalization in data-scarce settings. For example, SAM has been combined with feature interaction modules to enhance bi-temporal representation learning for change detection, as well as adapted to limited-data regimes through lightweight fine-tuning strategies [23], [24]. Although FireGAN does not directly employ foundation segmentation models, it aligns with this broader direction by leveraging strong cross-sensor feature learning to support downstream wildfire detection under limited training data.

To this end, we propose FireGAN, a GAN-based super-resolution framework specifically designed to enhance the spatial resolution of GOES ABI imagery for wildfire detection. By incorporating adversarial learning and perceptual loss, FireGAN addresses the limitations of existing AE-based models and achieves sharper reconstruction of fire boundaries, enabling more accurate and timely identification of wildfire activity.

III. Datasets and Study Domain

A. GOES and VIIRS Datasets

The ABI measurement aboard the GOES-18 satellite captures imagery across 16 channels with a temporal resolution of 10 min [25]. This high-frequency coverage enables consistent, near-real-time monitoring of the large-scale Western Hemisphere, making it particularly valuable for tracking rapidly changing events such as wildfires. Its 10-min gap in between imagery enables continuous monitoring of fire development and associated atmospheric changes, which is critical for forecasting and emergency response efforts.

TABLE I
Summary of Selected VIIRS and GOES ABI Channels Utilized for Wildfire Detection

Satellite Sensor	Central Wavelength (μm)	Spatial Resolution (m)	Repeat Cycle
VIIRS I04	3.74	375	16 days
VIIRS I05	11.45	375	16 days
GOES C07	3.90	2000	10 mins.
GOES C14	11.20	2000	10 mins.

Among the 16 spectral channels of the GOES ABI, channel 7, operating at 3.9 μm in the mid-wave infrared (MWIR) range, is especially well-suited for fire detection due to its strong sensitivity to thermal emissions from active fire zones [10]. It exhibits a rapid increase in spectral radiance with rising temperatures, making it highly responsive to fire-induced thermal anomalies [10], [26], [27]. In contrast, channel 14, centered at 11.3 μm in the longwave infrared (LWIR) range, is less sensitive to fire hotspots but is valuable for monitoring cloud-top temperatures and identifying smoke plumes [10], [26], [27]. When used together, channels 7 and 14 provide complementary information—channel 7 for identifying active fire pixels and channel 14 for contextual atmospheric features. These channels form the foundation of the GOES FDC product, which processes the observations from these channels every 10 min to identify and monitor fire activity. However, the native 2-km spatial resolution of both channels and consequently the FDC product limits the ability to detect small or early stage fires, reducing its effectiveness for fine-scale wildfire monitoring.

The VIIRS measurements onboard the Suomi-NPP, NOAA-20, and NOAA-21 satellites capture multispectral imagery across 22 channels [28]. Unlike GEO satellites, VIIRS operates from polar orbit and captures imagery twice with a 16-day revisiting gap for the same geographic locations. However, VIIRS offers significantly finer spatial resolution, with 375-m pixels in selected channels, making it well-suited for detailed fire analysis. This study focuses on channels I04 and I05, which have similar central wavelengths to GOES ABI channels 7 and 14, respectively. I04 and I05 provide detailed views of fire location, extent, and intensity. These channels are key components in the VIIRS VNP14, VJ114, and VJ214 fire products, which assign confidence levels (low, nominal, or high) to detected fire pixels based on observed thermal anomalies [11]. In this research, observations from VIIRS I04 and I05 serve as ground-truth references for training the FireGAN model. The characteristics of the selected ABI and VIIRS channels used in this study are summarized in Table I. In this study, GOES and VIIRS data from 2023 and 2024 are

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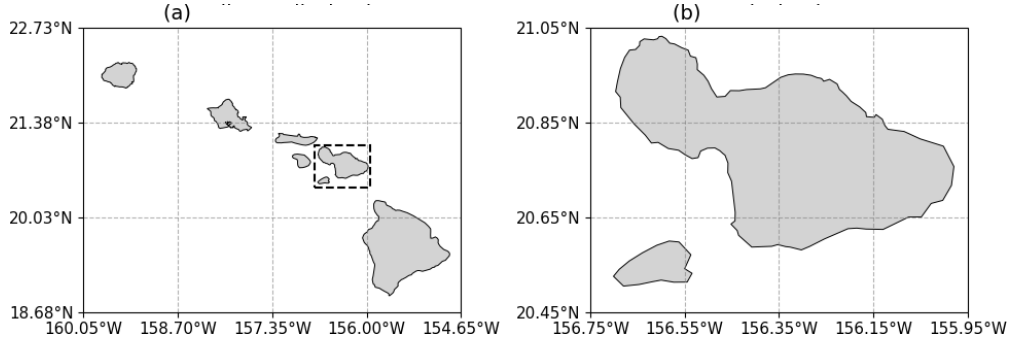


Fig. 2. Selected study domain in Hawaii: (a) full Hawaii Islands and (b) selected Maui Island highlighted by the black dashed box in (a).

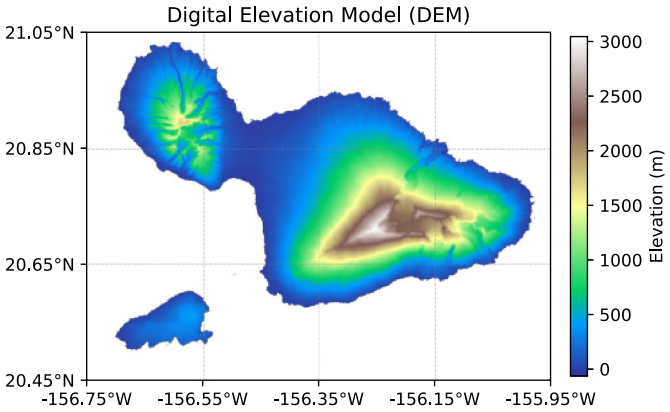


Fig. 3. Elevation map of Maui, HW shown in meters (m).

used for training and testing the FireGAN model, which was randomly divided into training and testing subsets.

B. Ground-Based Wildfire Boundary

To validate the satellite-derived wildfire detection results, a ground-truth boundary representing the total burn area from the August 2023 Maui wildfires was obtained from ground-based observations. This high-accuracy boundary was used as a reference to evaluate the spatial accuracy of the satellite-based wildfire perimeter estimates. By comparing the fire detection results generated from the downscaled GOES imagery against this detailed ground-based boundary dataset, the performance and reliability of the proposed fire detection approach can be assessed. Overlaying these maps with the ground-truth perimeter enables a direct assessment of how accurately the satellite-based system detects and localizes fire-affected areas.

C. Study Domain

This research focuses on the island of Maui, Hawaii, which is visualized in Fig. 2 as the study domain due to its combination of complex terrain, diverse ecosystems, and recent wildfire events. Maui contains pronounced geographical variability with elevations ranging from sea level to 3055 m (visualized in Fig. 3) at the peak of Haleakalā, a shield volcano and the island's highest point [29]. This large elevation range results in steep terrain gradients and heterogeneous land cover over short spatial scales, leading to high environmental complexity driven by both absolute elevation and elevation gradients. Geographically, the study domain is bounded by latitudes 20.45° N to 21.05° N and longitudes 156.75° W to 155.95° W. Maui's spatial extent is particularly well-suited for model development as it is large enough to yield diverse training data. Maui's varied elevation and land cover types present a challenging and representative environment for evaluating remote sensing techniques in wildfire detection. The August 2023 wildfires serve as a real-world case study, providing prominent thermal anomalies and satellite coverage for analysis.

IV. Methodology

A. Data Preprocessing

To ensure consistency between GOES and VIIRS datasets, radiance values from GOES and VIIRS are first converted to BTs for the selected channels. This conversion enables direct comparison across sensors in physically meaningful temperature units, facilitating accurate detection of thermal anomalies such as wildfires. To further isolate fire locations, we employ the active fire index (AFI) for GOES and VIIRS [30], respectively, as

$$\mathrm{AFI}_{\mathrm{GOES}} = \frac{\mathrm{BT}_{\mathrm{C07}} - \mathrm{BT}_{\mathrm{C14}}}{\mathrm{BT}_{\mathrm{C07}} + \mathrm{BT}_{\mathrm{C14}}} \tag{1}$$

where $\mathrm{BT}_{\mathrm{C07}}$ and $\mathrm{BT}_{\mathrm{C14}}$ are the BTs from GOES channels 7 and 14, respectively, and

$$\mathrm{AFI}_{\mathrm{VIIRS}} = \frac{\mathrm{BT}_{\mathrm{I04}} - \mathrm{BT}_{\mathrm{I05}}}{\mathrm{BT}_{\mathrm{I04}} + \mathrm{BT}_{\mathrm{I05}}} \tag{2}$$

where $\mathrm{BT}_{\mathrm{I04}}$ and $\mathrm{BT}_{\mathrm{I05}}$ are the BTs from VIIRS I04 and I05, respectively.

The AFI enhances the contrast between active fire pixels and the background by suppressing uniform temperature shifts and highlighting relative thermal anomalies [30]. High AFI values are indicative of intense fire activity, making the index a reliable metric in identifying wildfire hotspots. This approach is applicable to both GOES and VIIRS imagery, leveraging the sensitivity of MWIR channels to thermal emissions and the contextual information provided by LWIR measurements.

To accommodate the FireGAN architecture, the GOES imagery is resampled from 2 to 2.25 km, allowing a clean 6× spatial scaling to match the 375-m resolution of VIIRS data. After preprocessing, the GOES AFI and BTs from channels 7 and 14 have dimensions of $(27, 35)$, while the corresponding

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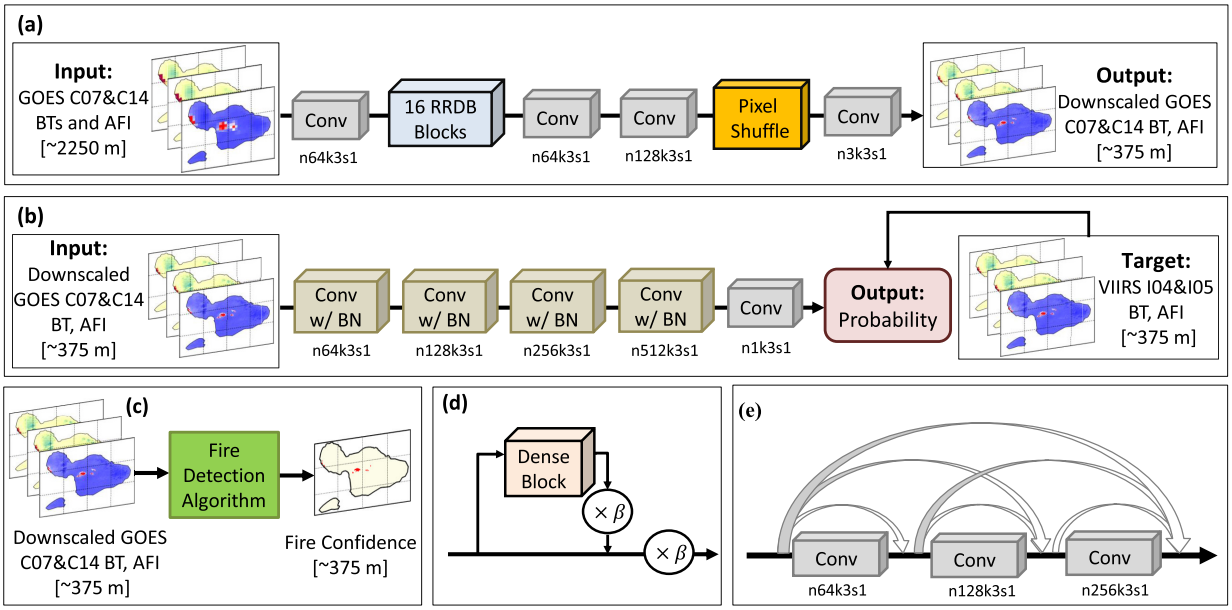


Fig. 4. Proposed framework for high-resolution wildfire detection. The FireGAN model consists of (a) generator and (b) discriminator. Details of the RRDB blocks and dense blocks are illustrated in (d) and (e), respectively. In the convolutional layers (Conv), n, k, and s represent the number of feature maps, kernel size, and stride size, respectively. The downscaled outputs from (a) generator are integrated into the fire detection algorithm in (c) to obtain the final fire detection confidence.

VIIRS AFI and BTs from I04 and I05 are of dimension $(162, 210)$.

To ensure data quality and temporal alignment between sources, the VIIRS imagery is paired with the temporally nearest GOES imagery over the selected study domain. Given the fixed 10-min temporal resolution of GOES ABI, the temporal offset between paired observations is bounded by ± 5 min. A data selection procedure is then applied to retain only high-quality image pairs for model training and evaluation. Specifically, VIIRS imagery is resampled to 2 km to match GOES spatial resolution, and correlation coefficients are computed for BTs and AFIs between the two datasets. Only image pairs with a correlation coefficient greater than 0.6 are retained, resulting in a high-quality dataset of 33 training and 26 testing scenes. Among the training set, there are 2 fire events in the training dataset and 1 fire event in the testing dataset. Prior to input into FireGAN, the BTs from both GOES and VIIRS are normalized between 0 and 1 to improve model convergence and mitigate potential biases during training.

B. FireGAN Model Architecture

GAN-based downscaling has demonstrated strong effectiveness in reconstructing fine spatial details from low-resolution inputs, making it well-suited for enhancing satellite imagery in environmental monitoring and wildfire detection applications [31], [32]. The proposed FireGAN consists of two networks as shown in Fig. 4: a generator $G(.)$ and a discriminator $D(.)$. As shown in Fig. 4(a), the generator is trained to produce high-resolution imagery from low-resolution GOES inputs, while the discriminator aims to distinguish the downscaled GOES imagery from the ground-truth VIIRS observations. These two networks are trained in an adversarial setting, wherein the generator is progressively encouraged to synthesize more realistic and perceptually accurate outputs [33], [34].

The generator in FireGAN is constructed using deep convolutional layers, 16 RRDBs, and a pixel-shuffle layer for upsampling. Each RRDB comprises a dense block that utilizes dense connections to facilitate feature reuse and improve gradient flow [35], [36]. Beginning with the RRDBs which are built from dense residual blocks (DRBs), each made up of interconnected convolutional layers, all of which have a linearly increasing number of feature maps from n64 $\rightarrow$ n128 $\rightarrow$ n256 as specified in Fig. 4(a), however they retain the same s (stride) of 1 along with the same k (kernel size) of 3. Each layer takes all previous outputs as input (dense connections), helping to preserve fine details and capture multiscale features. This captures a high range of hierarchies of features to retain features captured at each convolutional section. RRDBs compute multiple DRBs which are added together, allowing deeper and more powerful feature learning, in total the FireGAN uses 32 RRDBs. To stabilize training, a residual scaling parameter $\beta = 0.2$ is applied within each block. The pixel-shuffle layer increases the spatial resolution of the GOES imagery by a factor of 6, aligning it with the target 375-m resolution, followed by another convolutional layer to output final downscaled imagery. This architecture enables FireGAN to capture fine image details such as sharp fire boundaries and subtle thermal patterns by preserving both low- and high-frequency information throughout the network.

To distinguish between the downscaled GOES imagery and the corresponding ground-truth VIIRS observations, a discriminator network is trained, as illustrated in Fig. 4(b). The discriminator consists of a series of deep convolutional layers, each followed by a batch normalization (BN) layer, and concludes with a single output node using a

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sigmoid activation function to produce a probability score. rectified linear unit (ReLU) activations are used in all convolutional layers within both the generator and discriminator. In FireGAN, the discriminator loss function incorporates principles from the RaGAN framework [13], which refines the learning signal by considering the relative realism between real and generated samples. The discriminator is trained using a binary cross-entropy (BCE) loss that compares the generated super-resolved GOES imagery $y_f = G(x)$ from the generator, where x represents the low-resolution GOES input, against the high-resolution ground-truth VIIRS imagery y_r. The discriminator loss is defined as

$$\mathcal{L}_D := -\mathbb{E}_{y_r}\left[1 - \log D\left(y_f\right)\right] - \mathbb{E}_{y_f}\left[\log D\left(y_r\right)\right] \tag{3}$$

where $\mathbb{E}[.]$ denotes the expectation operator. $D(.)$ is the output probability from the discriminator indicating how closely the input resembles the ground-truth data. This loss formulation encourages the generator to produce outputs that are increasingly indistinguishable from true high-resolution VIIRS imagery.

The generator in FireGAN is optimized using a weighted combination of three loss components: adversarial loss $\mathcal{L}_{\text{Adv}}$, content loss $\mathcal{L}_{\text{Content}}$, and pixel-wise loss $\mathcal{L}_{\text{Pixel}}$. The total generator loss $\mathcal{L}_G$ is defined as

$$\mathcal{L}_G := \mathcal{L}_{\text{Content}} + \alpha_1 \cdot \mathcal{L}_{\text{Adv}} + \alpha_2 \cdot \mathcal{L}_{\text{Pixel}} \tag{4}$$

where the scaling factors $\alpha_1 = 0.3$ and $\alpha_2 = 150$ are empirically chosen to balance the contribution of each loss term, ensuring they remain on a comparable scale.

1) Content Loss: To move beyond pixel-wise similarity, FireGAN incorporates high-level perceptual features using a pretrained VGG-19 network [37]. Specifically, the content loss $\mathcal{L}_{\text{Content}}$ is defined as the mean squared error (mse) between deep feature representations extracted from layer 35 of VGG-19 for both the generated super-resolved output and the ground-truth VIIRS imagery. This enables the generator to capture structural and textural characteristics critical for preserving thermal gradients and accurately delineating fire boundaries—key requirements in wildfire detection tasks.

2) Adversarial Loss: The adversarial loss $\mathcal{L}_{\text{Adv}}$ is defined using the discriminator loss $\mathcal{L}_D$ described in (3). This encourages the generator to produce outputs that are indistinguishable from real high-resolution VIIRS imagery.

3) Pixel-Wise Loss: The content loss $\mathcal{L}_{\text{Content}}$ is computed as the mse between the generated imagery $y_f = G(x)$ and the ground-truth y_r, ensuring spatial accuracy by directly comparing pixel values.

FireGAN is trained in an adversarial manner, where the generator and discriminator are updated alternately. Both networks are optimized using the Adam optimizer [38], with gradients propagated through backpropagation [39]. This iterative training process enables the generator to progressively synthesize more realistic and structurally accurate super-resolved imagery, while the discriminator becomes increasingly adept at distinguishing generated outputs from real observations. The dynamic co-evolution of both networks ensures that the model converges toward producing high-quality, perceptually and numerically faithful wildfire imagery.

C. Fire Confidence Calculation

Following the proposed FireGAN, the GOES observations are downscaled to 375 m, enabling pixel-level computation of a fire confidence score used for final wildfire detection, as illustrated in Fig. 4(c). This confidence score serves as a probabilistic indicator of fire presence within the high-resolution imagery produced by FireGAN. We need to clarify here that the FireGAN component in our framework is used solely as a downscaling tool and does not perform any fire detection. The computation leverages key thermal features from the downscaled GOES data, including downscaled AFI_{GOES}, BT_{C07}, and BT_{C14} to identify likely fire-affected areas.

The resulting fire confidence score ranges from 0 to 1, with higher values indicating a greater likelihood of active fire presence. This score not only facilitates visual interpretation but also provides a quantitative metric for evaluating the model's predictions against ground-truth fire perimeters. The confidence scoring approach is inspired by established fire detection systems, including the GOES FDC product [10], the VIIRS active fire detection algorithm [11], and the MODIS active fire product [9]. As the primary contribution of our work is not the development of a new fire detection algorithm. We intentionally preserve the original detection logic from work in [11] with only minimal necessary adjustments. Our main focus is the downscaling of BT and AFI using the proposed GAN framework, after which the established fire detection algorithm is applied to generate the fire products. However, unlike these traditional methods, the proposed framework uniquely integrates the AFI, a feature that has demonstrated strong efficacy in highlighting thermal anomalies and improving hotspot detection in wildfire applications.

The pixel-wise fire confidence calculation begins with the definition of the normalization function to rescale input values to a bounded range

$$S\left(x, a, b\right) := clip\left(\frac{x-a}{b-a}, 0, 1\right) \tag{5}$$

where $clip(.)$ restricts the output to the range between 0 and 1, x represents the BT pixel value being normalized, and a and b are the upper and lower bounds for normalization.

To capture thermal anomalies indicative of fire activity, the BT difference between BT_{C07} and BT_{C14}, denoted as ΔT, is computed as

$$\Delta T := \text{BT}_{\text{C07}} - \text{BT}_{\text{C14}}. \tag{6}$$

Subsequently, let $\overline{\text{BT}}_{\text{C07}}$ and $\overline{\Delta T}$ be the mean of BT_{C07} and ΔT over all the pixels across the entire image, respectively, i.e.,

$$\overline{\text{BT}}_{\text{C07}} = \frac{1}{N}\sum_{i=n}^{N} \text{BT}_{\text{C07},n} \tag{7}$$

and

$$\overline{\Delta T} = \frac{1}{N}\sum_{i=n}^{N} \Delta T_n \tag{8}$$

where N is the total number of pixels in the BT imagery; $\text{BT}_{\text{C07},n}$ and ΔT_n refer to the nth pixel of BT_{C07} and ΔT, respectively. These statistical metrics serve as the foundation

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for subsequent robust standardization and anomaly detection steps used in the fire confidence estimation.

To robustly standardize thermal features and mitigate the influence of outliers, the median absolute deviation (MAD) is computed for $\mathrm{BT}_{\mathrm{C07}}$ and ΔT as

$$\mathrm{MAD}_{\mathrm{C07}} = \mathit{median}\left\{\left|\mathrm{BT}_{\mathrm{C07},n} - \overline{\mathrm{BT}}_{\mathrm{C07}}\right|\right\}_{n=1}^{N} \tag{9}$$

and

$$\mathrm{MAD}_{\Delta T} = \mathit{median}\left\{\left|\Delta T_n - \overline{\Delta T}\right|\right\}_{n=1}^{N} \tag{10}$$

respectively, where *median*(.) takes the median value of $\mathrm{BT}_{\mathrm{C07},n}$ or ΔT_n over all N pixels. These robust dispersion metrics are less sensitive to extreme values than standard deviation, making them more suitable for heterogeneous wildfire scenes.

Using these MAD values, standardized scores are computed for each pixel to normalize $\mathrm{BT}_{\mathrm{C07},n}$ and ΔT_n as

$$z_4\left(\mathrm{BT}_{\mathrm{C07},n}\right) = \frac{\mathrm{BT}_{\mathrm{C07},n} - \overline{\mathrm{BT}}_{\mathrm{C07}}}{\mathrm{MAD}_{\mathrm{C07}} + \epsilon} \tag{11}$$

$$z_{\Delta T}\left(\Delta T_n\right) = \frac{\Delta T_n - \overline{\Delta T}}{\mathrm{MAD}_{\Delta T} + \epsilon} \tag{12}$$

where ϵ is a small constant for numerical stability, which is set to $1\mathrm{e}^{-6}$ in this study. These standardized scores are used to identify thermal anomalies relative to the scene's background statistics, ensuring consistency across diverse conditions. Next, four normalized confidence components c_1 to c_4 are computed as intermediate scores for final fire confidence estimation

$$c_1 = \begin{cases} S\left(\mathrm{BT}_{\mathrm{C07}},\ 305,\ 320\right), & \text{if daytime} \\ S\left(\mathrm{BT}_{\mathrm{C07}},\ 305,\ 315\right), & \text{otherwise} \end{cases} \tag{13}$$

$$c_2 = S\left(z_4, 3, 6\right) \tag{14}$$

$$c_3 = S\left(z_{\Delta T}, 3.5, 6\right) \tag{15}$$

$$c_4 = \begin{cases} 1, & \text{if } \mathrm{AFI}_{\mathrm{GOES}} \geq 5 \\ \mathrm{AFI}_{\mathrm{GOES}}/5, & \text{otherwise.} \end{cases} \tag{16}$$

Here, c_1 accounts for absolute BT thresholds, c_2 and c_3 capture relative thermal deviations from scene means, and c_4 incorporates the AFI to emphasize fire-induced thermal patterns. These intermediate scores collectively support a robust and interpretable fire confidence framework. The final fire confidence score for each pixel is computed as the product of the four intermediate scores

$$\text{confidence} = c_1 \times c_2 \times c_3 \times c_4. \tag{17}$$

To ensure valid probability interpretation, the resulting confidence map is clipped to the range between 0 and 1. The output is a spatially continuous fire confidence field over the downscaled GOES imagery, where values closer to 1 denote a high likelihood of active fire presence. For interpretation and classification, pixels with confidence values greater than 0.75 are classified as high-confidence fire pixels, indicating a strong likelihood of active fire. Pixels with confidence values between 0.25 and 0.75 are treated as moderate-confidence regions, typically corresponding to fire perimeters, transition zones, or areas with partial fire activity due to mixed pixels or thermal diffusion effects. Scores below 0.25 are categorized as low-confidence, generally representing background or nonfire areas. This gradation provides not only pixel-level fire detection but also contextual awareness around active zones, thereby enhancing the operational utility of FireGAN outputs for early warning, response prioritization, and strategic wildfire management.

D. Evaluation Metrics

The proposed high-resolution fire detection framework is evaluated from two key perspectives: the effectiveness of the downscaling process and the accuracy of fire detection. All evaluations are conducted exclusively over land pixels within the designated study domain to ensure relevance and consistency. To assess the downscaling performance of FireGAN, we employ four widely used image quality metrics: root mean square error (RMSE), coefficient of determination (R^2), peak signal-to-noise ratio (PSNR), and structural similarity index measure (SSIM). These metrics jointly quantify pixel-level accuracy and perceptual similarity between the downscaled GOES imagery and the high-resolution VIIRS reference, capturing both numerical fidelity and visual quality.

RMSE quantifies the average pixel-wise difference between the super-resolved and ground-truth images

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum\nolimits_{n=1}^{N}\left(\hat{y}_n - y_n\right)^2} \tag{18}$$

where N denotes the total number of pixels; $\hat{y}_n$ and y_n are the nth pixel values from the downscaled and ground-truth images, respectively. Lower RMSE indicates better agreement with the reference. The coefficient of determination R^2 measures how well the predicted values approximate the actual values

$$R^2 = 1 - \frac{\sum_{n=1}^{N}\left(\hat{y}_n - y_n\right)^2}{\sum_{n=1}^{N}\left(\bar{y} - y_n\right)^2} \tag{19}$$

where $\bar{y}$ is the mean of the ground-truth pixel values. An R^2 closer to 1 implies better predictive accuracy. PSNR measures the ratio of the peak signal value to the mse

$$\mathrm{PSNR} = 10\log_{10}\left(\frac{\max_i\left\{y_i\right\}^2}{\frac{1}{N}\sum_{n=1}^{N}\left(\hat{y}_n - y_n\right)^2}\right) \tag{20}$$

where $\max_i\{y_i\}$ is the maximum pixel intensity in the ground-truth. Higher PSNR indicates lower distortion and better fidelity. Last, SSIM evaluates perceptual similarity by considering luminance, contrast, and structure

$$\mathrm{SSIM} = \frac{\left(2\mu_y\mu_{\hat{y}} + e_1\right)\left(2\sigma_{y\hat{y}} + e_2\right)}{\left(\mu_y^2 + \mu_{\hat{y}}^2 + e_1\right)\left(\sigma_y^2 + \sigma_{\hat{y}}^2 + e_2\right)} \tag{21}$$

where μ_y and $\mu_{\hat{y}}$ are the mean pixel values of the super-resolution and ground truth, respectively; σ_y^2 and $\sigma_{\hat{y}^2}$ are the variance of the super-resolution and ground truth, respectively; $\sigma_{y\hat{y}}$ is the covariance of the super-resolution and ground truth; the stability constants $e_1 = 6e - 4$ and $e_2 = 5e - 3$. SSIM values near 1 indicate high structural similarity. Together, these metrics offer a comprehensive and robust evaluation of FireGAN's downscaling quality.

To evaluate the fire detection performance of the proposed FireGAN framework, we employ several standard classification metrics: accuracy, precision, recall (also known as

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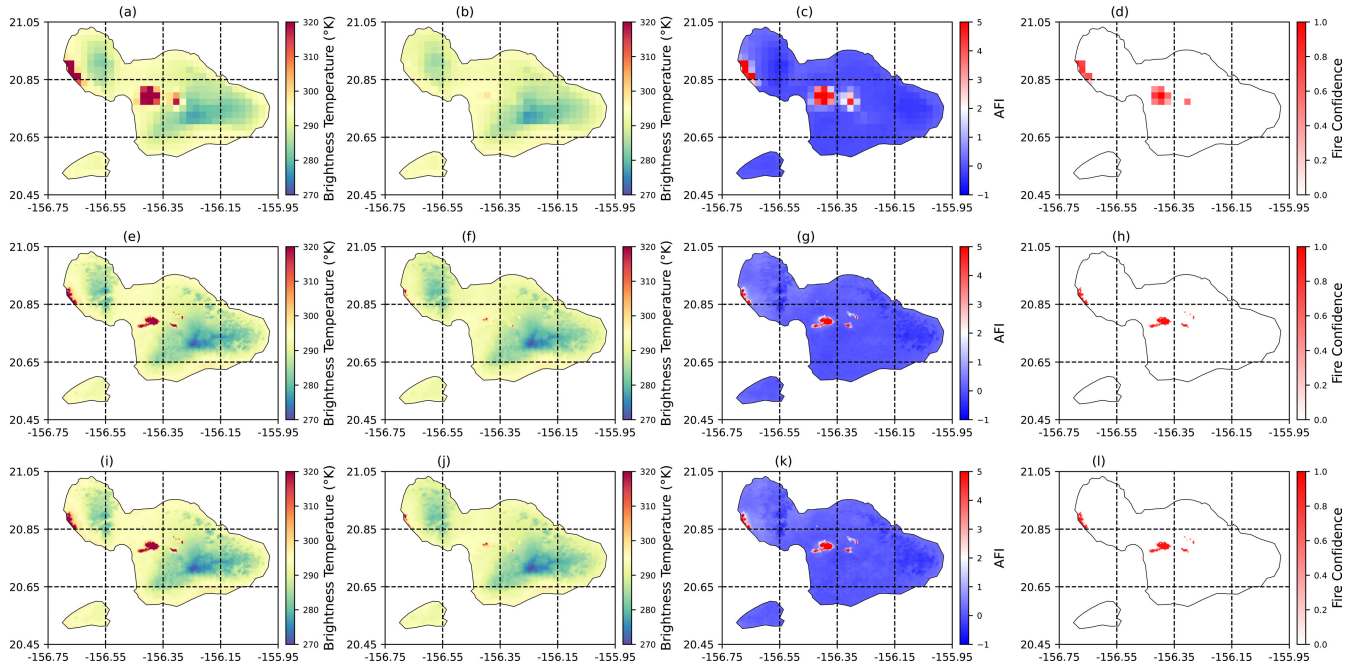


Fig. 5. Fire detection results at 12:12 UTC on August 9th, 2023: (a)–(d) low-resolution GOES observations and fire confidence from downscaled GOES data; (e)–(h) high-resolution VIIRS observations and fire confidence reference at 750-m resolution; and (i)–(l) downscaled GOES observations and fire confidence.

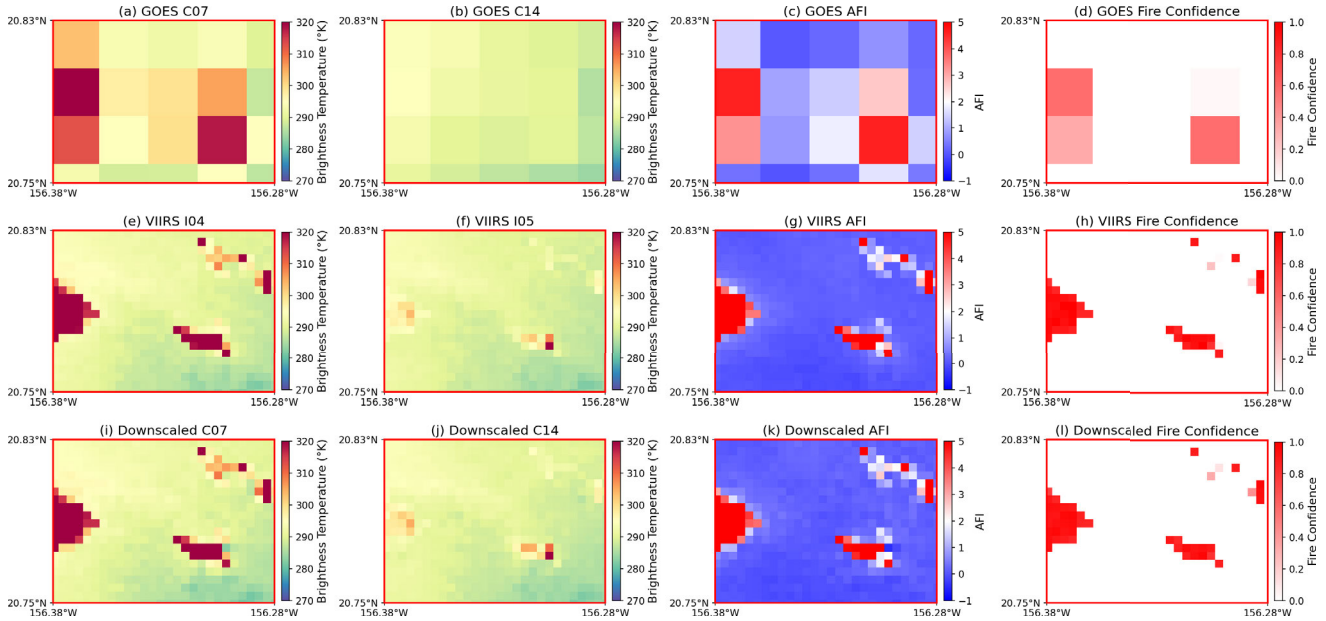


Fig. 6. Zoomed-in view of GOES-VIIRS, and downscaled GOES observations and fire confidence in Fig. 5 in Maui Island.

probability of detection), Heidke skill score (HSS), critical success index (CSI), and false alarm ratio (FAR). These metrics are defined as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{22}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{23}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{24}$$

$$\text{HSS} = \frac{2\,(\text{TP} \times \text{TN} - \text{FN} \times \text{FP})}{(\text{TP} + \text{FN})\,(\text{FN} + \text{TN}) + (\text{TP} + \text{FP})\,(\text{FP} + \text{TN})} \tag{25}$$

$$\text{CSI} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \tag{26}$$

$$\text{FAR} = \frac{\text{FP}}{\text{TN} + \text{FP}} \tag{27}$$

where true positive (TP) represents the number of pixels correctly predicted as fire; false positive (FP) refers to nonfire pixels incorrectly classified as fire; false negative (FN) is fire pixels incorrectly classified as nonfire; and true negative (TN) is nonfire pixels correctly classified. These metrics collectively assess both detection ability and reliability. Accuracy reflects the overall correctness, while precision and recall

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TABLE II
EVALUATION METRICS BASED ON ALL THE TESTING DATASET

Metric	Acc.	Precision	Recall	HSS	CSI	FAR
Score	0.998	0.849	0.829	0.839	0.724	0.148

provide insights into the correctness of fire predictions and the system's sensitivity to actual fires, respectively. HSS accounts for random chance and is commonly used to evaluate model skill in environmental monitoring. CSI provides a balanced evaluation of detection and false alarms, especially in imbalanced data scenarios. FAR quantifies the proportion of false alarms among all predicted fire cases. In general, higher values of recall, HSS, and CSI, coupled with lower FAR, indicate superior fire detection performance.

V. EXPERIMENTAL RESULTS

A. Experimental Setups

Before the training process, all input and target images were normalized using min–max scaling from 0 to 1. During inference, the outputs were denormalized to recover the original physical units for interpretation and evaluation. The FireGAN model was trained for 24 000 epochs, allowing both the generator and discriminator to adequately learn complex spatial transformations from low-resolution GOES inputs to high-resolution VIIRS labels. This extensive training schedule was aligned with prior work in GAN-based super-resolution, where prolonged adversarial training was critical for generating perceptually realistic and structurally accurate images [32], [34]. The Adam optimizer with a learning rate of $1e^{-6}$ was used to update model weights for both networks.

B. Wildfire Detection Results

Fig. 5 presents an example of the fire confidence map generated by FireGAN during the August 2023 wildfire event in the testing set, at a spatial resolution of 375 m using downscaled GOES imagery. To validate results, the fire confidence output was resampled to 750 m to match the spatial resolution of the VIIRS active fire product. A visual comparison across these subfigures in Fig. 5 and the zoomed-in view in Fig. 6 indicates that the proposed framework demonstrates strong agreement with the VNP14 product, suggesting the effectiveness and accuracy of FireGAN in high-resolution wildfire detection.

To quantitatively assess the wildfire detection performance, the fire confidence results from FireGAN were resampled from 375 to 750 m to align with the spatial resolution of the VIIRS active fire product, which was used as the reference due to the absence of a consistent high-resolution active fire dataset. In the evaluation, pixels with fire confidence scores exceeding 0.1 were classified as fire pixels. The performance metrics are summarized in Table II. As shown in the table, the proposed framework achieved a high overall accuracy of 0.998, indicating strong agreement with the VIIRS active fire product. A precision of 0.849 suggests that 85% of the fire detections made by the model were correct. The recall score of 0.829 demonstrates the model's effectiveness in identifying actual fire events. Furthermore, the HSS of 0.839 and a CSI of 0.724 confirmed that the model performs well beyond random chance. The low FAR further emphasized the reliability of the detections. Collectively, these results highlighted the effectiveness of the proposed high-resolution wildfire detection framework in accurately identifying wildfire activity from super-resolved satellite imagery.

TABLE III
EVALUATION METRICS ON THE TESTING DATASET

Metric	Score $\pm$ Standard Deviation
RMSE	0.240 ± 0.163
R^2	0.999 ± 0.001
PSNR	29.252 ± 9.916
SSIM	0.966 ± 0.027

TABLE IV
PERFORMANCE SCORES AT 12:10 UTC ON NOVEMBER 2, 2023

Channel	RMSE	R^2	PSNR	SSIM
C07	0.324	0.999	36.057	0.986
C14	0.269	0.999	38.045	0.991
AFI	0.014	0.999	17.385	0.969

C. FireGAN Downscaling Results

Table III summarizes the average quantitative evaluation of the FireGAN downscaling capability over all testing samples. The low RMSE indicates a small pixel-wise error between the super-resolved outputs and the ground-truth, suggesting a high level of pixel-wise accuracy. R^2, with a value approaching 1, suggested that the model captures the majority of the variance in the high-resolution data, demonstrating strong predictive capability. The PSNR value of 29.252 dB indicates that the generated high-resolution images have a good signal quality, although the relatively high standard deviation reflects variability across different test samples. Finally, the SSIM score of 0.966 signified a strong structural resemblance between the generated images and the ground-truth, further validating FireGAN's capability in preserving image details during downscaling. Overall, these metrics confirmed that FireGAN effectively enhances the spatial resolution of GOES imagery with high accuracy and structural consistency.

1) Nonfire Case: Fig. 7 illustrates a downscaling example for a no-fire case at 12:10 UTC on November 2, 2023, drawn from the testing set, along with the corresponding downscaling performance metrics in Table IV. This example highlighted the ability of FireGAN to accurately downscale GOES imagery with sufficient texture details and patterns compared to the ground-truth VIIRS observations. The GOES channel 7 BT achieved strong performance with an RMSE of 0.324, a PSNR of 36.057, and an SSIM of 0.986, suggesting effective noise suppression and high perceptual quality. Channel 14 BT exhibited the best results, with a lower RMSE of 0.269 and a high PSNR of 38.045, reflecting minimal error to the VIIRS reference. The AFI demonstrated a notably low

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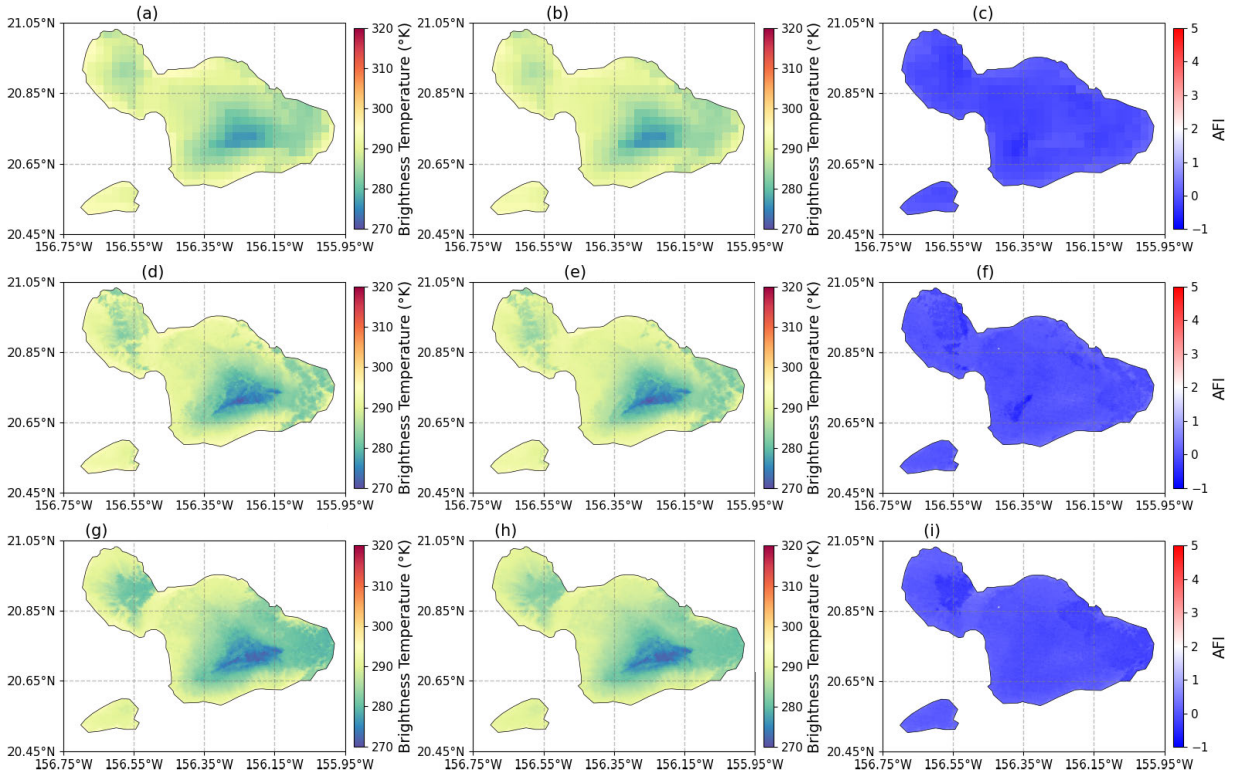


Fig. 7. Downscaling results for a no-fire case on November 24th, 2023, at 12:10 UTC. (a)–(c) GOES AFI, and channels 7 and 14 BTs. (d)–(f) VIIRS AFI, I04 and I05 BTs. (g)–(i), downscaled GOES AFI, and channels 7 and 14 BTs.

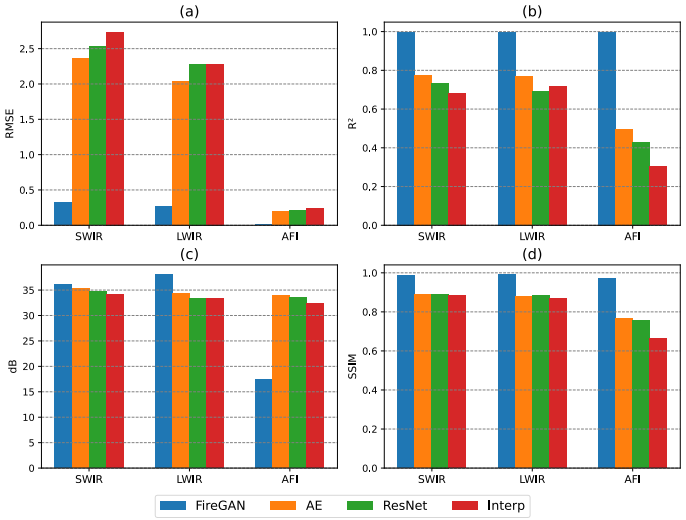


Fig. 8. Bar plot comparison of DL model performances, including FireGAN (blue), auto encoder (orange), ResNet (green), and interpolation (red). (a) RMSE. (b) R^2. (c) PSNR. (d) SSIM.

RMSE of 0.014 and a high SSIM of 0.969, reflecting strong structural consistency. Although the AFI's PSNR was lower at 17.385, its significantly reduced RMSE showed the model's strength in preserving fine-scale thermal anomalies critical for hotspot identification. Across all channels, the R^2 values were consistently 0.999, confirming a near-perfect correlation between FireGAN outputs and ground-truth data. In addition, we conducted experiments using an AE [14] and a residual network (ResNet) [36] as benchmark models for comparison, along with bicubic interpolation. Both models were trained on the same dataset as FireGAN. The resulting downscaled products are summarized in Fig. 8. As shown, VegGAN achieves substantially lower RMSE than the competing methods in both the SWIR and LWIR channels, demonstrating the superiority of our FireGAN-based downscaling framework. Together, these metrics validated the model's ability to deliver high-fidelity, high-resolution reconstructions under nonfire conditions.

2) Fire Case: This section presents a fire case study of FireGAN's performance during the August 9, 2023, wildfire event on the island of Maui, Hawaii. The analysis spanned a 12-h period, from 03:00 UTC through 14:00 UTC, with

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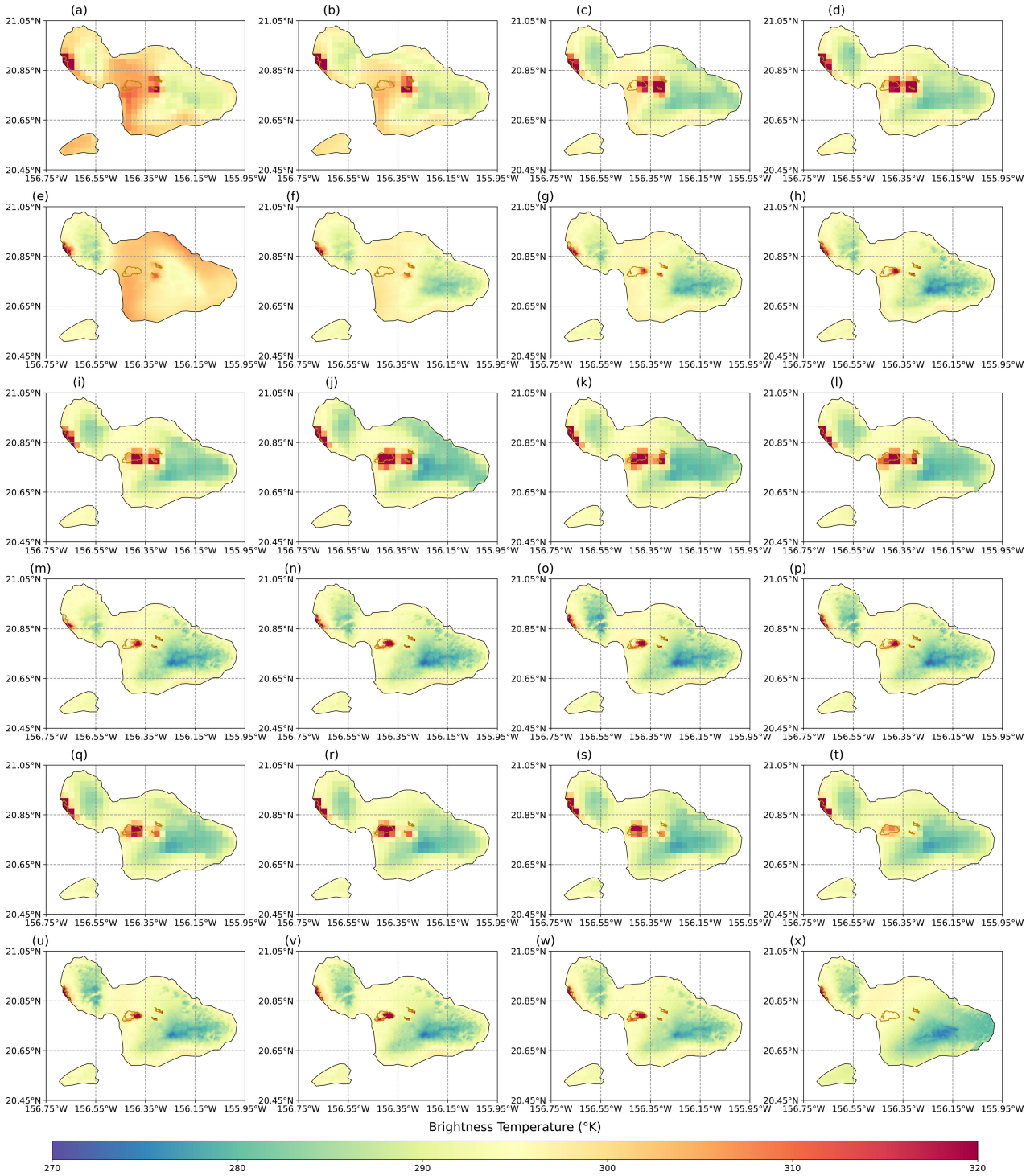


Fig. 9. Consecutive downscaling results of GOES channel 7 BT with 1-hour interval on a fire case over Maui island, HI, from 03:00 to 14:00 UTC on August 9th, 2023. The orange contour depicts the ground-based wildfire boundary. The 1st, 3rd, and 5th rows are before downscaling, while the 2nd, 4th, and 6th rows are after downscaling.

satellite imagery captured at hourly intervals. The input and super-resolved outputs generated by FireGAN are shown in Figs. 9–11, for channels 7 and 14 BTs, AFI, and fire confidence, respectively. These images were overlaid with the ground-truth wildfire perimeter obtained from daily field observations, enabling temporal tracking of fire progression across the island.

TABLE V

FIRE DETECTION METRICS AT 12:10 UTC ON AUGUST 9TH, 2023 ACROSS OTHER DL MODELS AND INDERPOLATION

Model	Accuracy	Precision	Recall	FAR	CSI	HSS
FireGAN	0.998	0.849	0.829	0.148	0.724	0.839
AE	0.992	0.422	0.657	0.578	0.346	0.510
ResNet	0.992	0.421	0.621	0.579	0.335	0.498
Interpolation	0.993	0.425	0.594	0.574	0.330	0.492

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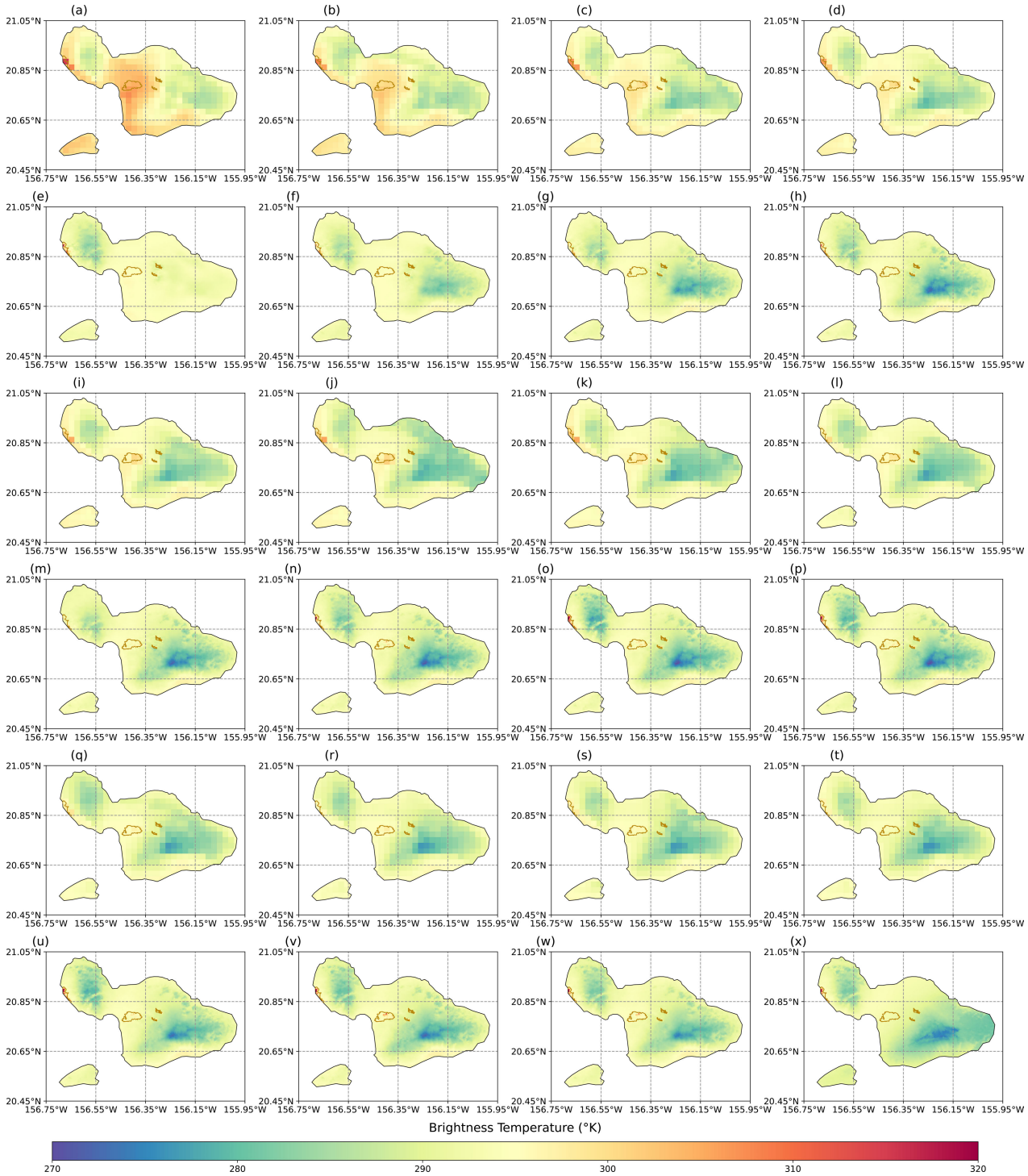


Fig. 10. (a)–(x) Same as Fig. 9 but for downscaled GOES channel 14 BTs.

The full-island FireGAN results revealed active fire regions that were difficult to distinguish at the native 2-km resolution of the original GOES ABI imagery. By enhancing the spatial resolution to 375 m, FireGAN provided significantly clearer delineation of fire zones, enabling more accurate spatial analysis. These downscaled outputs also formed the basis for focused evaluations of fire-affected subregions. Importantly, the fire confidence maps derived from the super-resolved AFI outputs closely matched the observed ground-based fire boundaries in Fig. 12, demonstrating FireGAN's ability to detect and track active fires with high spatial precision. The temporal evolution captured over the 12-h window clearly illustrated the wildfire's growth and spatial dynamics. Notably, while elevated AFI and channel 14 BT values were observed around 03:00 UTC in certain areas, the fire confidence remained low outside the established fire perimeter, indicating that FireGAN effectively suppresses false positives by incorporating multichannel context into its confidence calculation.

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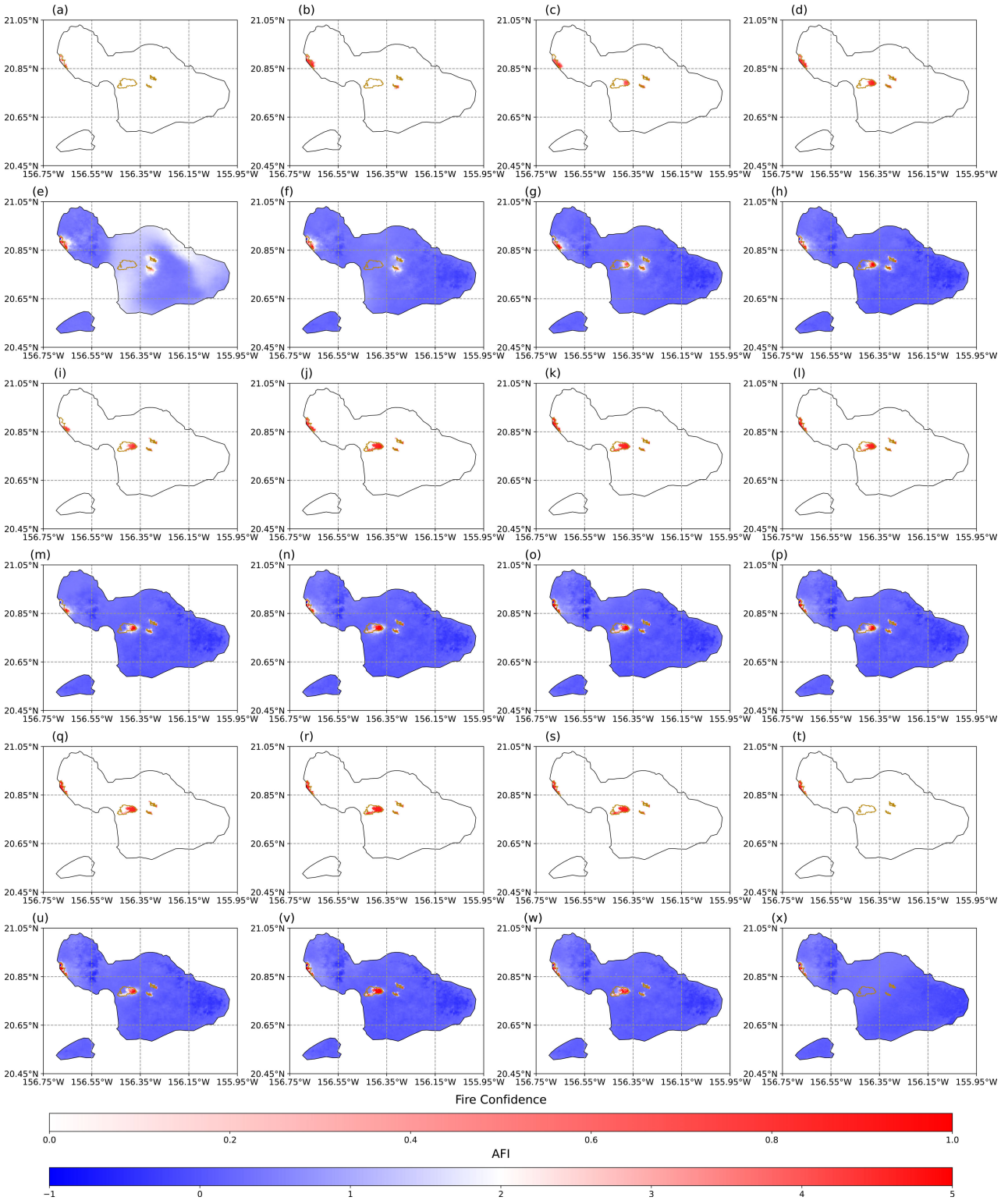


Fig. 11. (a)–(x) Same as Fig. 9 but for downscaled GOES fire detection and AFI.

To further assess the downscaling quality and fire detection accuracy, we performed a detailed analysis by zooming into two subregions shown in Fig. 13 in the following and quantitative evaluation. Table V presents the quantitative evaluation of this fire event, together with results from the AE, ResNet, and bicubic interpolation. As shown in Table V, FireGAN consistently outperforms all other methods across the listed metrics, as FireGAN particularly had higher precision, CSI, and HSS, and much lower FAR compared to AE, ResNet, and interpolation.

a) Lahiana fire: The zoomed-in region focused on the town of Lahaina, Maui, which experienced some of the most severe wildfire damage during the August 2023 event. This localized analysis enables a more detailed evaluation

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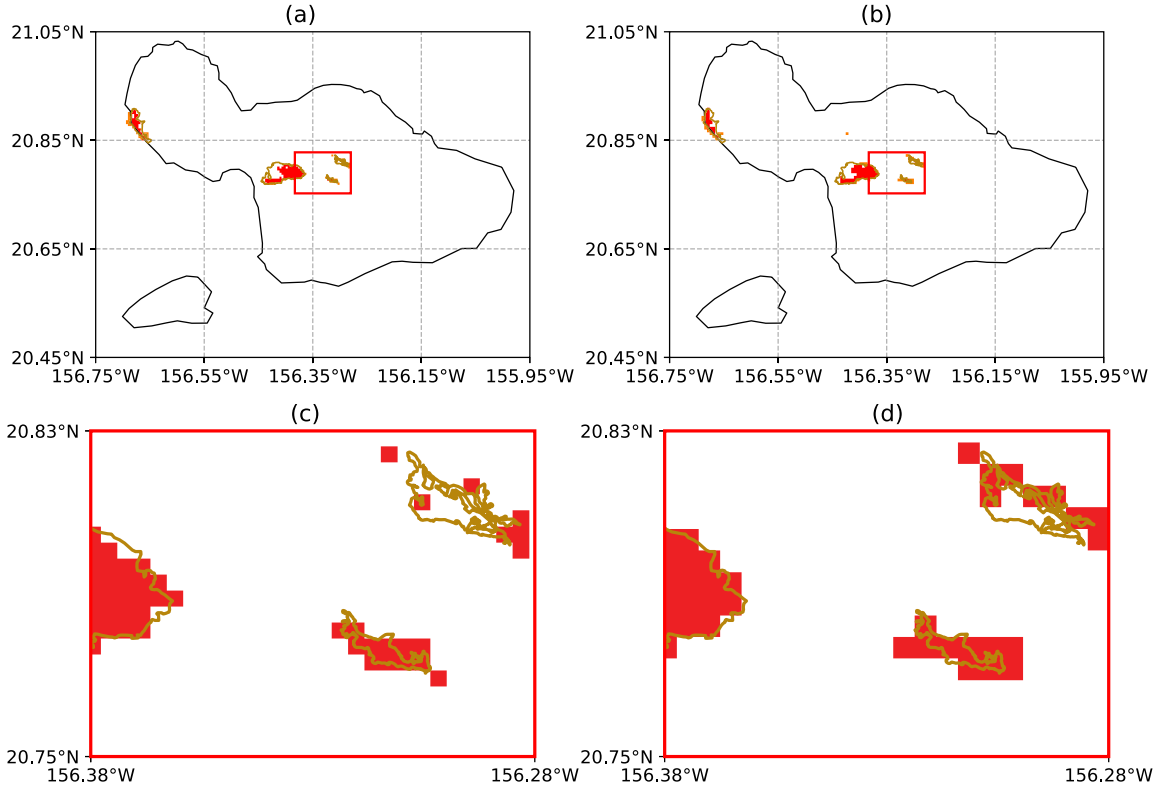


Fig. 12. Comparison of the derived fire detection results on August 9th, 2023, with the ground-based fire boundary. (a) Derived 375-m fire confidence from FireGAN; (b) VIIRS VNP14 fire product; and (c) and (d) are zoomed-in views of (a) and (b) in the red box region, respectively. Correctly detected fire pixels are shown in red, while false alarms are shown in orange.

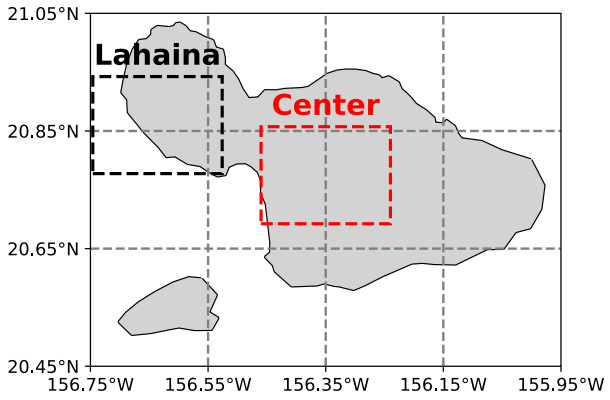


Fig. 13. Boxed Lahaina along with the center region shown to reference the two zoomed-in locations.

of FireGAN's performance in both super-resolving thermal features and detecting fire activity.

The zoomed-in results for the Lahaina fire, presented in Fig. 14, clearly illustrate the temporal evolution and spatial accuracy of FireGAN's fire detection outputs from 03:00 UTC to 14:00 UTC. The super-resolved AFI imagery, overlaid with the ground-based wildfire boundary, revealed strong alignment between high-confidence fire pixels and the actual burn perimeter. Notably, at 03:00 and 04:00 UTC, a cluster of false positive fire pixels appeared outside the established fire boundary. However, the associated confidence values for these pixels diminished steadily over subsequent hours, falling below 0.3, which indicates the model's ability to self-correct. This refinement in confidence underscores the robustness of the FireGAN framework in mitigating false detections while maintaining sensitivity to true fire activity.

Within the fire boundary, areas exhibiting moderate confidence values between 0.2 and 0.6 may correspond to the regions where the fire is either emerging or subsiding, potentially representing partially burning pixels. These transitional zones can be critical in capturing the dynamic behavior of the wildfire front. Such information was highly valuable for emergency responders, offering insights into the fire's progression and enabling more informed decisions for localized containment, resource allocation, and tactical planning.

b) Center region fire: The zoomed-in portion of central Maui Island highlighted the Pulehu, Olinda, and Kula wildfires—collectively referred to as the center fires in this study to represent the broader fire-affected region depicted in Fig. 13. This classification facilitated regional analysis and tracking of fire behavior across multiple ignition points within the central zone of the island.

Fig. 15 shows the fire confidence and downscaled AFI for the center region of Maui Island, spanning from 03:00 to 14:00 UTC. The fire confidence maps effectively mitigated many of the false positives observed in earlier outputs, underscoring the robustness of the confidence calculation in accurately identifying fire hotspots. Notably, the edges of the fire zones displayed lower confidence values, which were indicative of fire expansion or dissipation over time. This behavior was particularly evident between 05:00 and 09:00 UTC, during which the Pulehu fire exhibited rapid growth. Initially detected with moderate confidence values around 0.35, the affected area

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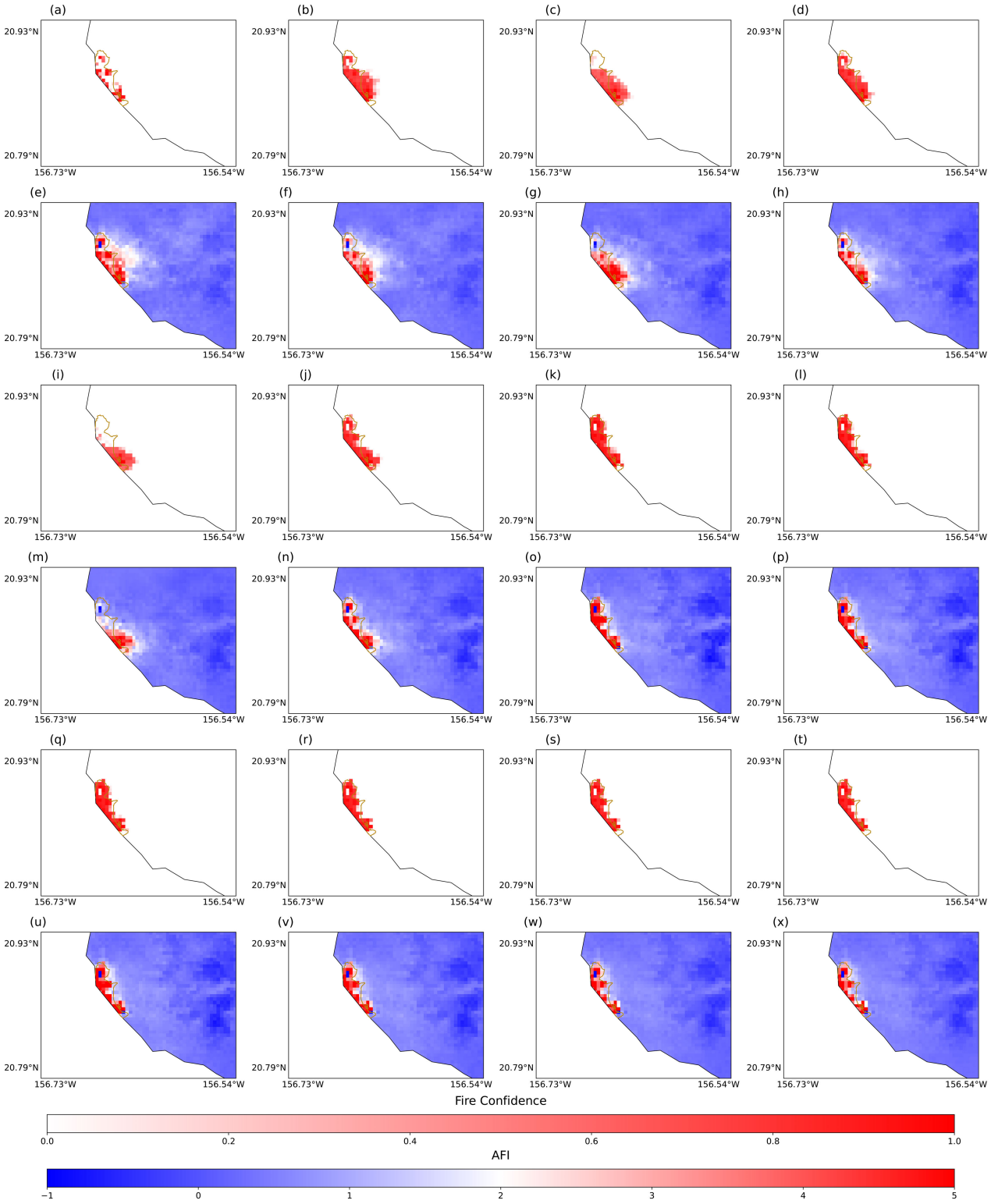


Fig. 14. (a)–(x) Detailed fire confidence and downscaled AFI for Lahiana of Maui, HI from 03:00 to 14:00 UTC on August 9th, 2023.

intensified and expanded, culminating in a well-established and prominent fire hotspot in subsequent hours.

VI. Discussion

A. Evaluation of Wildfire Detection Results

In this study, the performance of the proposed wildfire detection framework was evaluated primarily using limited scenes from the VIIRS fire product. Although VIIRS provides high spatial resolution, its infrequent revisit rate over Maui Island results in a relatively small number of usable scenes for robust validation. On the other hand, products such as the GOES FDC algorithm offer high temporal frequency but lack the spatial resolution required for detailed benchmarking. Therefore, at the time of this research, no existing fire

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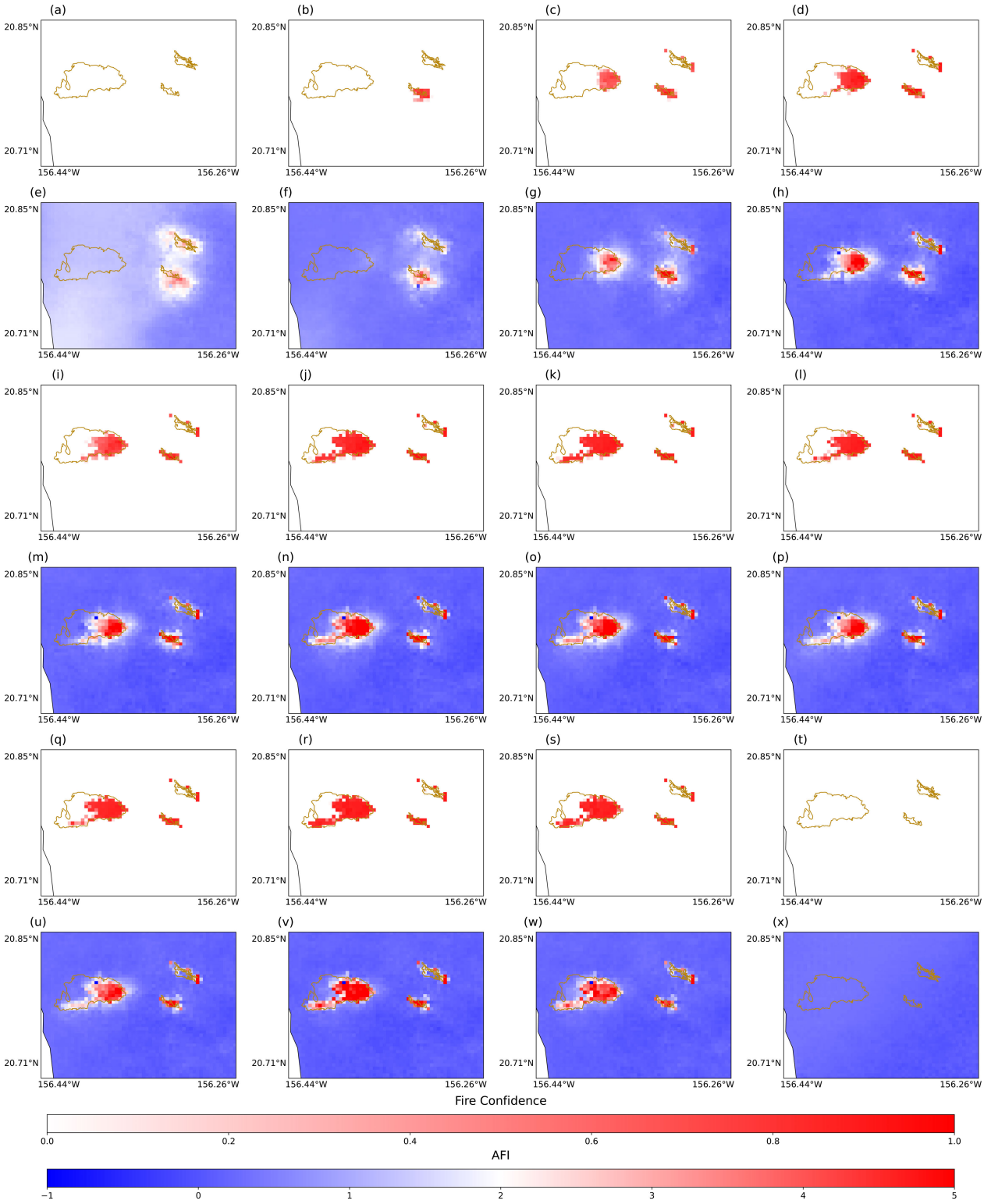


Fig. 15. (a)–(x) Detailed fire confidence and downscaled AFI for the center region of Maui, HI from 03:00 to 14:00 UTC on August 9th, 2023, including Pulehu, Olinda, and Kula fires.

detection product provided both the spatial and temporal resolution necessary to serve as a fully reliable reference. In addition, the evaluation relies on paired GOES-VIIRS observations that may exhibit a small temporal mismatch due to differences in satellite overpass timing. Such a mismatch can introduce uncertainty arising from rapid-fire evolution or cloud and smoke motion. However, the temporal offset between paired observations is bounded by the GOES sampling interval and reflects realistic operational conditions under which geostationary and polar-orbiting data are jointly used. As a result, these effects are unlikely to dominate the learned spatial downscaling relationship or materially bias the evaluation.

This limitation shows the need for advanced solutions such as FireGAN, which aim to bridge the spatial–temporal

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resolution gap by generating high-resolution fire detection outputs while preserving the high temporal frequency of geostationary observations.

B. Generalization of FireGAN

In this work, FireGAN was trained exclusively over Maui Island, Hawaii. To extend its applicability to other geographic regions or different satellite datasets, domain adaptation poses a significant challenge. One promising direction to enhance the generalizability of FireGAN is the incorporation of continual learning and transfer learning techniques [40], [41]. These approaches would allow the well-trained FireGAN model to adapt to new domains with minimal retraining, thereby enabling its deployment in other wildfire-prone regions, such as California [42]. This forms a critical component of our future work. By downscaling GOES ABI observations across different regions, FireGAN could generate active fire products with both high temporal and high spatial resolution, ultimately enhancing wildfire detection and situational awareness in diverse environments.

C. Active Wildfire Detection Algorithm

In this work, we adopted active fire detection algorithms from GOES, VIIRS, and MODIS systems as a foundation for our study region [9], [10], [11]. A key novelty of our approach is the incorporation of the AFI into the fire detection process. This enhancement led to fire detection results that are both reliable and promising, showing strong agreement with the VIIRS active fire product and ground-based wildfire boundaries. However, it is important to note that the parameters used in our fire detection algorithm may require adjustment when applied to other regions or different environmental contexts. In addition, the primary focus of this research is on developing a novel downscaling framework (FireGAN) to enhance GOES measurements for fire detection. Therefore, the active fire detection module implemented in this work may not represent the optimal detection strategy. Recent advances in DL-based fire detection methods using satellite imagery have demonstrated improved performance and adaptability (see [43], [44]). Integrating such data-driven approaches into our framework could further improve detection accuracy in operational settings and is considered a valuable direction for future work.

VII. Conclusion

Despite the availability of various satellite-based fire products, there remains a strong demand for fire detection solutions that offer both high spatial and temporal resolutions. This gap significantly limits the operational utility of current satellite-based active wildfire detection systems. For instance, while the GOES satellite provides observations every 10 min—making it advantageous for near-real-time monitoring—its coarse 2-km spatial resolution lacks the detail necessary for accurate fine-scale wildfire detection.

To address this limitation, this research introduces FireGAN, a GAN-based framework designed to enhance the spatial resolution of fire-related GOES imagery. The model is trained using high-resolution (375 m) VIIRS observations as supervisory labels over the island of Maui, Hawaii. Experimental results demonstrate the effectiveness of FireGAN in accurately downscaling satellite imagery in both fire and nonfire scenarios. Importantly, FireGAN enables the generation of enhanced wildfire detection products from GOES ABI data with a high temporal resolution (10 min) and improved spatial resolution (375 m).

While this research establishes a promising foundation for super-resolving wildfire imagery, several avenues remain for future enhancement. One potential improvement is the integration of additional data sources or channels. Incorporating auxiliary information from other GOES ABI channels or complementary datasets may help improve detection robustness and accuracy. Another promising direction involves a hierarchical downscaling approach—first enhancing GOES imagery using FireGAN and subsequently aligning it with even higher resolution datasets such as Landsat or Sentinel. This multistage enhancement could yield extremely detailed fire mapping, offering further benefits for wildfire monitoring, response, and mitigation.

Acknowledgment

The GOES and VIIRS data used in this research are available from the Amazon Web Services (AWS) and LAADS DAAC at https://noaa-goes18.s3.amazonaws.com/index.html and https://ladsweb.modaps.eosdis.nasa.gov/, respectively. The intermediate features in the FireGAN model and the model parameters are available upon request. The scientific results and conclusions, as well as any views or opinions expressed herein, are those of the authors and do not necessarily reflect those of the United States Department of Agriculture.

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